Mango Materials

Company Overview Mango Materials is a California-based biotechnology company that produces biodegradable bio-polymers. They use a unique natural process to turn waste methane gas—a major contributor to climate change—into eco-friendly bioplastics that can replace petroleum-based plastics in packaging and textiles.

ESG Focus: Methane-to-Bioplastic (PHA Polymers)

Why (The Problem) Methane is a greenhouse gas that is over 28 times more potent at trapping heat in the atmosphere than carbon dioxide. At the same time, the world is flooded with traditional plastics made from oil, which take hundreds of years to break down and heavily pollute our oceans and landfills.

How (The Solution) Mango Materials uses a special type of bacteria called "methanotrophs." They feed these bacteria waste methane gas collected from wastewater treatment plants, landfills, or agricultural facilities. The bacteria digest the methane and naturally produce a biopolymer called PHA (polyhydroxyalkanoate) inside their cells as an energy reserve.

What (The Outcome) The company extracts this PHA and turns it into pellets that can be used to make biodegradable packaging, films, and fibers. This results in a material that behaves like traditional plastic but is completely bio-based. If a Mango Materials product ends up in the ocean, marine microbes can eat it, causing it to safely biodegrade in weeks.

Impact: This technology creates a double-positive impact: it prevents dangerous methane gas from entering the atmosphere and replaces harmful oil-based plastics. They are leading the way toward a truly circular bio-economy where waste is used as a valuable raw material.